

The DRanking Protocol: Verifiable Distributed Ranking and Sybil-Bounded Admission for Structured Peer-to-Peer Networks

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Abstract

DRanking converts locally observed peer behavior in an open structured peer-to-peer network into a globally agreed, verifiable service weight that gates admission to protocol roles. Identity creation is free, so no mechanism bounds adversarial count [1]; the protocol instead bounds adversarial weight and prices it. The instruments are: counterparty-signed service receipts in place of unsigned counters; a seed-anchored trust flow $t_{k+1} = \alpha C_k^T t_k + (1 - \alpha)s$ obeying conservation, attenuation, and revocation laws, with a proved bound $T_k(A) \leq \alpha g / (1 - \alpha)$ on Sybil mass; trust-weighted verifiable random sortition in place of oracle sampling; a proof-of-authority ledger trusted for ordering only, with equivocation proofs as first-class slash objects; and a circuit-friendly epoch transition folded by Nova-family incrementally verifiable computation, giving constant-cost verification to browser nodes. Rank is soulbound and slashable; reward is a separate transferable asset; the authority-to-stake transition is conditioned on a security-budget crossover.

I. Introduction

The 2023 draft of this protocol scored behavior per identity. Identities are free; every component secured against lying nodes fell to numerous nodes. This rewrite keeps the draft's decomposition — local evidence, global aggregation, sampling, ledger, incentives — and replaces the quantity being secured:

$$\begin{aligned} \text{cost}(\text{did}) &= 0, \\ |\text{Adv} \cap V| &\leq \infty, \\ \text{secure}(| \cdot |) &= \perp \quad [1], \\ \text{secure} &:= \text{secure}(T(\cdot)), \\ T(A) &= \sum_{v \in A} t(v), \\ \text{claim} &= \text{price}(T(\text{Adv}) \geq \tfrac{1}{3}). \end{aligned}$$

Sections II–VIII construct t , its evidence, its sampling, its ledger, and its economy; Section IX tabulates the attack surface. Each mechanism is a term of price.

II. Model

Definition 1 (Substrate).

$$\begin{aligned} R &= \mathbb{Z}/2^{160}\mathbb{Z} \quad [2], [3], \\ \text{did} &: K \rightarrow R, \quad \text{self-certifying}, \\ \text{link} &\subseteq V \times V, \quad \text{authenticated}, \\ \text{msg} &= \langle m, \sigma_{\text{origin}} \rangle, \\ \text{cost}(\text{did}) &= 0, \\ \text{cost}(\text{link}) &= \varepsilon > 0. \end{aligned}$$

Definition 2 (Epoch).

$$k \in \mathbb{N}, \quad \text{epoch}(k) = [\text{fin}(\Sigma_k), \text{fin}(\Sigma_{k+1})),$$

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delimited by ledger finality (Definition 9).

Assumption 1 (Adversary).

$$\begin{aligned} |\text{Adv}| &\leq \infty, \\ \text{collude, grind, patience} &\in \text{Cap}(\text{Adv}), \\ \text{byzantine} &\in \text{Cap}(\text{Adv}), \\ \text{forge}(\sigma_h), h \notin \text{Adv} &\notin \text{Cap}(\text{Adv}), \\ \text{break}(\text{VRF}) &\notin \text{Cap}(\text{Adv}). \end{aligned}$$

Assumption 2 (Honest weight).

$$T_k(\text{Adv}) < \tfrac{1}{3} T_k(V) \quad \forall k.$$

No bound is assumed on $|\text{Adv}|$.

III. Service Receipts

Unsigned counters are first-person testimony; aggregated, they are hearsay. The unit of evidence is a counter-signed receipt.

Definition 3 (Receipt).

$$\begin{aligned} \rho &= \langle \text{srv}, a, b, k, n, \sigma_b \rangle, \\ \text{srv} &\in \{\text{relay, store, probe}\}, \\ a &= \text{server}, \quad b = \text{signer/served}, \\ \text{valid}(\rho) &\equiv V(k_b, \sigma_b, \text{srv} \| a \| b \| k \| n) = \top \\ &\quad \wedge (b, k, n) \text{ fresh}. \end{aligned}$$

Proposition 1 (Forgery surface).

$$\text{valid}(\rho) \wedge b \notin \text{Adv} \Rightarrow \rho \text{ issued by } b.$$

Proof. Immediate from $\text{forge}(\sigma_b) \notin \text{Cap}(\text{Adv})$ (Assumption 1). Manufacturable receipts are exactly those with $b \in \text{Adv}$ (self-dealing); these are neutralized by signer-trust weighting (Section IV). $\square$

Definition 4 (Receipt graph). □

$$\begin{aligned} W_k &\in \mathbb{R}_{\geq 0}^{V \times V}, \\ W_k[b, a] &= \sum \{\text{val}(\rho) : \rho \text{ valid, sig } b, \text{ srv } a, \text{ ep } k\}, \\ W_k[b, \cdot] \mathbf{1} &\leq w_{\max}. \end{aligned}$$

The cap makes countersigning a budgeted act; the protocol certifies performed service, never synthetic load (cf. Constraint 1).

IV. Trust Algebra

Definition 5 (Seed vector).

$$\begin{aligned} s &\in [0, 1]^V, \quad \sum_v s_v = 1, \\ \text{supp}(s) &= S \quad (\text{authority set}). \end{aligned}$$

Law 1 (Conservation).

$$C_k[b, a] = \frac{W_k[b, a]}{\max(w_{\max}, W_k[b, \cdot] \mathbf{1})}, \quad C_k[b, \cdot] \mathbf{1} \leq 1.$$

Law 2 (Attenuation).

$$\alpha \in (0, 1), \quad \text{flow}(\ell \text{ hops}) \leq \alpha^\ell.$$

Law 3 (Revocation).

$$\begin{aligned} \text{slash}(v) &\Rightarrow t(v) := 0, \quad v \notin \text{carrier}(t), \\ t_{k+1} &= f(W_k, t_k) \quad (\text{re-derived per epoch}), \\ s &\quad \text{withdrawable per seed.} \end{aligned}$$

Definition 6 (Trust update).

$$t_{k+1} = \alpha C_k^\top t_k + (1 - \alpha) s. \quad (1)$$

One power-iteration step per epoch toward $t^* = (1 - \alpha)(I - \alpha C^\top)^{-1} s$ [4], [5]; the fixpoint is approached along chain time, never computed within an epoch (cf. Definition 11).

Invariant 1 (Mass).

$$\sum_v t_0(v) = 1 \Rightarrow \sum_v t_k(v) \leq 1 \quad \forall k.$$

Proof.

$$\begin{aligned} \sum_v t_{k+1}(v) &= \alpha \mathbf{1}^\top C_k^\top t_k + (1 - \alpha) \\ &\leq \alpha \mathbf{1}^\top t_k + (1 - \alpha) \quad (\text{Law 1}) \\ &\leq \alpha + (1 - \alpha) = 1 \quad (\text{induction}). \end{aligned}$$

□

Proposition 2 (Sybil influence bound). Partition $V = H \uplus A$, $S \subseteq H$. Let

$$g_k = \sum_{b \in H, a \in A} C_k[b, a] t_k(b), \quad g = \sup_k g_k.$$

Then

$$T_k(A) = \sum_{v \in A} t_k(v) \leq \frac{\alpha g}{1 - \alpha} \quad \forall k.$$

Proof. $S \cap A = \emptyset$ kills the seed term over A . Splitting (1) by origin region and applying Law 1 to internal circulation,

$$T_{k+1}(A) \leq \alpha(g_k + T_k(A)).$$

From $T_0(A) = 0$,

$$T_k(A) \leq \alpha g \sum_{i \geq 0} \alpha^i = \frac{\alpha g}{1 - \alpha}.$$

$T(A)$ is linear in honest endorsement granted (g), independent of $|A|$. Defense concentrates on g : endorsement spends budget (Law 1), and detection claws it back (Law 3).

Definition 7 (Soulbound rank).

$$\begin{aligned} \text{rank}_k(v) &= t_k(v), \\ \text{transfer} &\notin \text{Ops}(\text{rank}), \\ \text{rank} &\xrightarrow{\text{slash}} 0, \\ \text{rank} &\xrightarrow{\text{epoch}} \text{re-derived (1)}. \end{aligned}$$

(Cf. Proposition 7.)

Definition 8 (Dilution schedule).

$$\begin{aligned} \text{vol}_k &= \sum_{i \leq k} \mathbf{1}^\top W_i \mathbf{1}, \\ 1 - \alpha_k &= D(\text{vol}_k), \quad D \text{ non-increasing, published,} \\ t_0 &= s \quad (\text{cold start}). \end{aligned}$$

V. Sortition

Proposition 3 (Statistical undetectability). Model $\text{did} = H(\text{pk})$, H a random oracle. For the adversarial placement strategy “sample fresh keys,” honest and adversarial positions are identically distributed on R :

$$\Delta_{\text{TV}}(\text{pos}(H \cup A), \text{pos}(H')) = 0 \quad \forall |A|,$$

hence every position-based test (entropy, Kolmogorov–Smirnov) has distinguishing advantage 0 against Sybil density.

Proof. $H(\cdot)$ uniform on R for every key, honest or adversarial; a union of i.i.d. uniforms is distribution-invariant in its partition. □

Placement tests detect biased sampling; the threat is cheap identity. Validation must be cryptographic and economic, and a global random seed is a bootstrap circularity (a randomness beacon presupposes the committee being selected). Both are removed by sortition [6], [7].

Algorithm 1 Trust-weighted sortition (node v , epoch k , role r)

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- 1: $B_k \leftarrow \text{hash}(\text{fin}(\Sigma_k))$ $\triangleright$ beacon: fixed by finality
 - 2: $(y, \pi) \leftarrow \text{VRF}_{\text{sk}_v}(B_k \| r)$
 - 3: if $y/2^{|y|} < \tau \cdot t_k(v)$ then
 - 4: v holds r in epoch k ; publishes π
-

Proposition 4 (Grinding closure).

$$\begin{aligned} \text{sk}_v \text{ fixed} &\prec B_k \text{ exists,} \\ B_k \text{ fixed} &\prec \text{Adv acts,} \end{aligned} \Rightarrow \text{grind} = \perp.$$

Proof. Both VRF inputs are committed before either party can condition on the output; unpredictability from Assumption 1. □

Proposition 5 (Committee safety). Let seats be independent indicators with $\mathbb{E}[X] = \tau$, $\mathbb{E}[X_A] = \tau T(A)$, $T(A) < \frac{1}{3}$. By Chernoff–Hoeffding,

$$\Pr\left[X_A \geq \frac{\tau}{3}\right] \leq \exp\left(-\tau D\left(\frac{1}{3} \parallel T(A)\right)\right),$$

with D the binary relative entropy: committee dishonesty decays exponentially in committee size. Sortition is the transport of Assumption 2 from weight to seats; no reported quantity exists for a coalition to steer, so the draft’s median game is vacuous by construction.

VI. Admission Ledger

Definition 9 (Ledger).

$$\begin{aligned} \Sigma_k &= \langle \text{rt}(t_k), \text{rt}(V_k), \text{rt}(\text{sl}_k), \text{rt}(\text{bal}_k), \text{ver} \rangle, \\ \text{fin}(\Sigma_k) &\equiv \text{QC}_{2/3}(S_k) \quad [8], [9], \\ \text{content} &= \{\text{rt}(W_k)\} \cup \text{sl} \cup \Delta V \cup \text{issue}, \\ \text{data}(W_k) &\in \text{Overlay storage (not on chain)}. \end{aligned}$$

Obligation 1 (Minimal trust).

$$\begin{aligned} \forall o \in \text{content} : \text{Verify}(o) &\perp \text{honest}(S), \\ \text{Cap}(\text{quorum}) &= \{\text{order}, \text{censor}\}, \\ \text{forgeable}(\text{quorum}) \ni o &\Rightarrow \text{design defect}. \end{aligned}$$

Definition 10 (Equivocation proof).

$$\begin{aligned} e &= \langle m_1, \sigma_v(m_1), m_2, \sigma_v(m_2) \rangle, \\ \text{slot}(m_1) = \text{slot}(m_2) \wedge m_1 \neq m_2 &\Rightarrow \text{slash}(v), \end{aligned}$$

verification assumption-free; e is a first-class ledger object, submittable by anyone.

Obligation 2 (Force inclusion).

$$\text{ack}_S(x) \wedge x \notin \Sigma_{k+\Delta} \Rightarrow \text{ack}_S(x) \in \text{sl (quorum slash)},$$

bounding censorship, the residual validator power, by latency Δ .

Equivocation defeats patience: months of farmed standing forfeit on one contradiction, so $\mathbb{E}[\text{farm}] \downarrow$ with detection probability.

VII. Folded State

Browser wasm nodes can neither replay history nor audit a validator signature; the chain must be succinct [10].

Definition 11 (Epoch transition).

$$F(\Sigma_k, \text{rt}(W_k)) = \Sigma_{k+1},$$

Algorithm 2 F — pure, deterministic, per epoch

- 1: $\text{sl}_{k+1} \leftarrow \text{sl}_k \cup \{\text{verify}(e) : e \text{ submitted}\}$
 - 2: $t' \leftarrow t_k \upharpoonright_{V \setminus \text{sl}_{k+1}}$ ▷ Law 3
 - 3: $t_{k+1} \leftarrow \alpha_k C_k^T t' + (1 - \alpha_k) s$ ▷ Eq. (1), Def. 8
 - 4: $\text{bal}_{k+1} \leftarrow \text{bal}_k + \text{issue}(W_k, t_k) - \text{burn}(\text{fees}_k)$
 - 5: $V_{k+1} \leftarrow \text{vset}(\Sigma_{k+1})$ ▷ Obligation 6
 - 6: return $\langle \text{rt}(t_{k+1}), \text{rt}(V_{k+1}), \text{rt}(\text{sl}_{k+1}), \text{rt}(\text{bal}_{k+1}), \text{ver} \rangle$
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$\text{cost}(F) = O(\text{nnz}(W_k))$ adds/mults over committed data.

Line 3 is why the fixpoint is unrolled: one sparse matrix–vector product arithmetizes trivially; an eigenvector solve does not.

Definition 12 (Folding). Nova-family IVC [11]:

$$\begin{aligned} \pi_k &\vdash \Sigma_0 \xrightarrow{F^k} \Sigma_k, \\ |\pi_k| &= O(1), \quad \text{Verify}(\pi_k) = O(1), \\ \text{sync} &= \text{Verify}(\pi_k) \wedge \text{QC}_{2/3}(\Sigma_k), \\ \text{prove} &\parallel \text{consensus (asynchronous, trails head)}. \end{aligned}$$

Proposition 6 (Linearity).

$$\text{fin}(\Sigma_k) \text{ instant (BFT)} \Rightarrow \text{chain} \cong (\Sigma_0, \Sigma_1, \dots) \text{ linear,}$$

hence IVC applies with no re-folding branch; probabilistic-finality chains admit no such reduction.

Proof. Quorum intersection: two finalized blocks at one height require $\geq \frac{1}{3}$ equivocating signers, contradicting Assumption 2 via Definition 10. $\square$

Obligation 3 (Circuit discipline).

$$\begin{aligned} \forall \text{ extension } F' : \\ F' \text{ pure} \wedge F' \in \text{CheapArith}(\text{committed}). \end{aligned}$$

The ledger is an admission ledger, not a contract platform.

Obligation 4 (Upgrade versioning).

$$\begin{aligned} F \mapsto F' &\Rightarrow \text{vk} \mapsto \text{vk}', \\ \text{ver} &\in \Sigma, \\ \text{bind}(\text{vk}, \text{vk}') &\text{ at committed epoch; } \end{aligned}$$

shipped at genesis, not retrofitted.

Folding provides validity only: **canonicity** $\in$ **consensus**, **availability** $\in$ **overlay**, **privacy**(W_k) open (granularity of $\text{rt}(W_k)$: full graph, aggregates, or blinded commitments — a standing design decision with surveillance consequences).

VIII. Economics

Definition 13 (Two assets).

$$\begin{aligned} \text{rank} &: \text{soulbound (Def. 7),} \\ &\quad \text{gates admission/sortition,} \\ \text{reward} &: \text{transferable,} \\ \text{issue}_k(v) &\propto (W_k^T t_k)(v). \end{aligned}$$

Proposition 7 (Payable rank defeats cost anchoring). Let rank be transferable at price p . Then

$$\begin{aligned} \text{cost}_{\text{Adv}}(m) &= \min(\text{farm}(m), p \cdot m), \\ u_{\text{exit}}(\text{hold}) = 0 \wedge p > 0 &\Rightarrow \text{exiting nodes sell,} \\ \text{supply} > 0 &\Rightarrow p \leq \inf_v \text{farm}_v(m)/m, \end{aligned}$$

so acquisition cost decouples from behavior. Non-transferability removes the second branch of the min, restoring $\text{farm}(m) = \text{service} + \text{time} + \text{slash risk}$ as the sole path. $\square$

Definition 14 (Fee loop).

$$\begin{aligned} \text{Sinks} &= \{\text{store}_{>p}, \text{QoS}, \text{namespace}\}, \\ \text{fees} &\rightarrow \text{burn} \cup \text{issue pool}, \\ \text{anchor}_{\text{PoS}} &\leq f(\text{demand}(\text{Sinks})). \end{aligned}$$

Obligation 5 (Security-budget crossover).

$$\text{activate(PoS)} \iff \text{Bond} \geq \beta \cdot \widehat{\text{payoff}}(\text{attack}), \quad \beta > 1,$$

conditioned, never scheduled; pre-crossover, PoA finality overlays PoS production, weighted down by $D(\text{vol}_k)$ (Definition 8).

Obligation 6 (Validator-set abstraction).

$$\begin{aligned} \text{vset} &: \Sigma \rightarrow 2^V, \\ \text{vset}_{\text{PoA}} &= \text{const } S, \\ \text{vset}_{\text{PoS}} &= \text{sortition}(\text{Bond}); \end{aligned}$$

the transition changes one pure function, not the engine.

Constraint 1 (Rejected anchors).

$$\begin{aligned} \text{PoW} &: \text{cost}(51\%) = \text{rent}(\text{hash}) \cdot O(\text{hours}) \Rightarrow \perp, \\ \text{PoT} &: W_{\text{synth}}[b, a], \quad a, b \in \text{Adv} \not\subseteq W_{\text{real}} \\ &\Rightarrow \text{bandwidth burned}, \Delta T(\text{Adv}) \approx 0 \Rightarrow \perp. \end{aligned}$$

Receipted real service with trust weighting is proof-of-traffic minus the synthetic load.

IX. Security Analysis

Table I
Threat matrix

Attack	Bound	Residual
Sybil flood	Prop. 2	$\alpha g / (1 - \alpha)$
Self-dealing	Prop. 1 + signer-trust weight; cap (Def. 4)	zero-trust circulation
Patient farming	Laws 1–3; total slash (Def. 10)	undetected deviation
Whitewashing	fresh DID $\Rightarrow t = 0$; slow earn, instant loss	re-entry at zero
Grind/eclipse	Prop. 4	—
Censorship	Obl. 2	latency Δ
Seed compromise	threshold S ; revocation; Def. 8	pre-dilution window
Rank market	Prop. 7	out-of-band key sale

$$\begin{aligned} \text{sale}(\text{sk}_v) &\Rightarrow \text{buyer inherits slash exposure}, \\ \text{PoA} &= \text{trust root until Obl. 5.} \end{aligned}$$

X. Related Work

Local scoring descends from eDonkey queues [12] and BitTorrent choking [13]; propagation from EigenTrust and personalized PageRank [4], [5], here unrolled along epochs over signed evidence; the attack-edge bound pattern from SybilGuard/SybilLimit/SybilRank [14]–[16], instantiated over a service graph. Brahms [17] addresses adversarial sampling that the draft’s tests could not (Proposition 3); DRanking sidesteps via Algorand-style sortition [6], [7] weighted by trust. S/Kademlia [18] prices identifier creation (complementary one-time cost). Finality follows PBFT lineage [8], [9]; succinctness follows Mina [10] via Nova folding [11]; impossibility framing is Douceur’s [1]; the external-cost anchor observation is Nakamoto’s [19].

XI. Conclusion

$$\text{safe} \iff \text{cost}(T(\text{Adv}) \geq \tfrac{1}{3}) > \widehat{\text{payoff}}(\text{attack}),$$

and every mechanism above — receipts, the trust laws, sortition, the ledger, folding, the two-asset economy — is a term on the left-hand side.

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